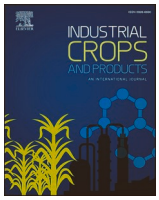
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One-step extraction and hydrolysis of crocin into crocetin by recyclable deep eutectic solvents

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ABSTRACT

Crocetin, an aglycone derived from crocin, is synthesized in biological systems through the hydrolysis of crocin as a bioactive metabolite, which is closely associated with its pharmacodynamic effects. This research demonstrates, for the first time, that deep eutectic solvents (DES) could not only exhibit excellent solubilizing properties for crocin-based compounds, but also offer a protic medium capable of catalyzing the cleavage of glycosides to yield crocetin. A series of bifunctional and recyclable DES was proposed and characterized using attenuated total reflection Fourier-transform infrared spectroscopy (ATR-FTIR), differential scanning calorimeter (DSC), and cytotoxicity analysis. The benzyltriethylammonium chloride-oxalic acid dihydrate (TEBAC-OA, 1:1.5, mol/mol) was tailor-made for the simultaneous extraction and conversion process with a solid-to-liquid ratio of 1:20 and heating at 80 °C for 30 min. The yield of crocetin could reach 8.485 mg/g. Following the processing, DES and crocetin were successfully recovered separately, resulting in a 94% recovery rate for crocetin. The used DES was then utilized in the subsequent processing step, maintaining a production efficiency of 95.8%. Additionally, the cytotoxicity assessment revealed that the final used eutectic systems exhibited lower cytotoxicity compared to other commonly used solvents. The hydrogen bonding energy between DES and crocin, as well as the acidity of DES, were found to have a strong correlation with the yield of crocetin. Consequently, this study has successfully developed a sustainable, efficient, and simple technique for the one-step production of crocetin from gardenia fruit.

1. Introduction

Gardenia jasminoides Ellis, a flowering plant belonging to the Gardenia genus and Rubiaceae family, is cultivated throughout China and utilized in traditional medicine for the treatment of various conditions such as inflammation, viral encephalitis, hepatitis, tonsillitis, tracheitis, and high fever (Zhang et al., 2007). Crocin, a water-soluble carotenoid ester derived from crocetin, is primarily found in *Crocus sativus* L. and *Gardenia jasminoides* Ellis (Pfister et al., 1996), and offers numerous health benefits (Sheng et al., 2006; Tao et al., 2014; Zhou et al., 2015). However, in 2005, Asai et al. made the discovery that crocetin and its glucuronide conjugates were detectable in plasma after the oral administration of crocin, but not crocins (Asai et al., 2005). Thus, it can be inferred that crocins undergo metabolism to crocetin either prior to or during intestinal absorption (Xi et al., 2007). Crocetin, a constituent of crocin, tricrocin, dicrocin, and picrocrocin, has been identified as a significant contributor to the antioxidant process in an antioxidant activity test (Ochiai et al., 2007). There are notable distinctions between crocin and crocetin in terms of their nutritional, nutraceutical, and pharmacological benefits, which can be attributed to their different chemical structures (Fiorito et al., 2022). The structures of crocetin and crocin are shown in Fig. 1(A). Notably, crocetin demonstrated the highest inhibitory effect on lipid peroxidation compared to crocin and completely revealed remarkable super antioxidant activity (Chen et al., 2008). Crocetin exhibited various mechanisms of action, including enhancement of oxygen transport and diffusivity, suppression of pro-inflammatory mediators, cellular protection against reactive oxygen species (ROS) damage, and induction of apoptosis in cancer cells (Guo et al., 2022). Additionally, crocetin may possess other pharmacological properties, such as antihypertensive and antithrombotic effects (Higashino et al., 2014), alleviating renal dysfunction (Tang et al., 2006), inhibiting retinal damage (Ishizuka et al., 2013), improving the quality of sleep (Kuratsune et al., 2010), and preventing insulin resistance (Xi et al., 2005), making it an area of promising research for obtaining pure crocetin or crocetin-enriched blends from natural plant sources.

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The conventional procedure of crocetin production (from *Gardenia jasminoides* Ellis and *Crocus sativus* L.) entailed initial extraction of crocin using an organic solvent, followed by sequential treatment with alkali/acid or enzyme for a specified duration, and subsequent extraction with ethyl acetate to acquire crude crocetin (Reddy et al., 2020). This approach was time-intensive and involved the utilization of volatile and hazardous (for operators) acids/bases, along with substantial quantities of organic solvents. Qian et al. employed Amberlite D140 resin chromatography as a method of extracting gardenia yellow pigment from gardenia fruit (Qian et al., 2010). Subsequently, the obtained materials underwent alkali-hydrolysis followed by acidification. Ultimately, crocetin was crystallized from dimethylformamide. Lautenschläger et al. utilized two distinct enzyme preparations, namely RöhmEnzym® and Rohament CL®, to carry out enzymatic deglycosylation for the production of crocetin from *Crocus sativus* L. (Saffron) (Lautenschläger et al., 2014). Recently, a noteworthy approach has emerged that allows to produce crocetin without the need for alkalization and acidification steps. Fiorito et al. conducted an evaluation on a panel consisting of 21 synthetic clays, of which 4 exhibited selective retention of crocetin in the solid phase after hydrolysis of crocin (Fiorito et al., 2022). The resulting suspension was subsequently subjected to stirring for a duration of 96 h before undergoing filtration and adsorption. This process also demonstrated a multistep, discontinuous, time-consuming, and labor-intensive approach. Furthermore, it is worth noting that Saffron, known as red plant gold, is a costly resource (approximately $1000/kg) due to limited availability and low production. In contrast, *Gardenia jasminoides* Ellis is a more affordable option (about $1/kg) and widely cultivated (Zhang et al., 2007).

The development of sustainable solvents is a critical issue in green chemistry. Deep eutectic solvents (DES) were first introduced in 2003 by Abbott and co-workers (Abbott et al., 2003a, 2003b). Because of their unique properties such as non-toxicity, biodegradability, cheap cost, solute stability, and ease of preparation, the use of DES as a green solvent in the production of valuable compounds has grown in favor in recent years (Bi et al., 2020). In comparison to conventional organic solvents and volatile organic compounds derived from non-renewable sources, DES have demonstrated effective utilization in plant extraction, offering numerous advantages such as enhanced efficiency, reduced time requirements, cost-effectiveness, non-toxicity, biodegradability, and improved product purity (Ling and Hadinoto, 2022). Synthetic organic solvents possess favorable extraction capabilities contingent upon their polarity; however, DES, comprising hydrogen

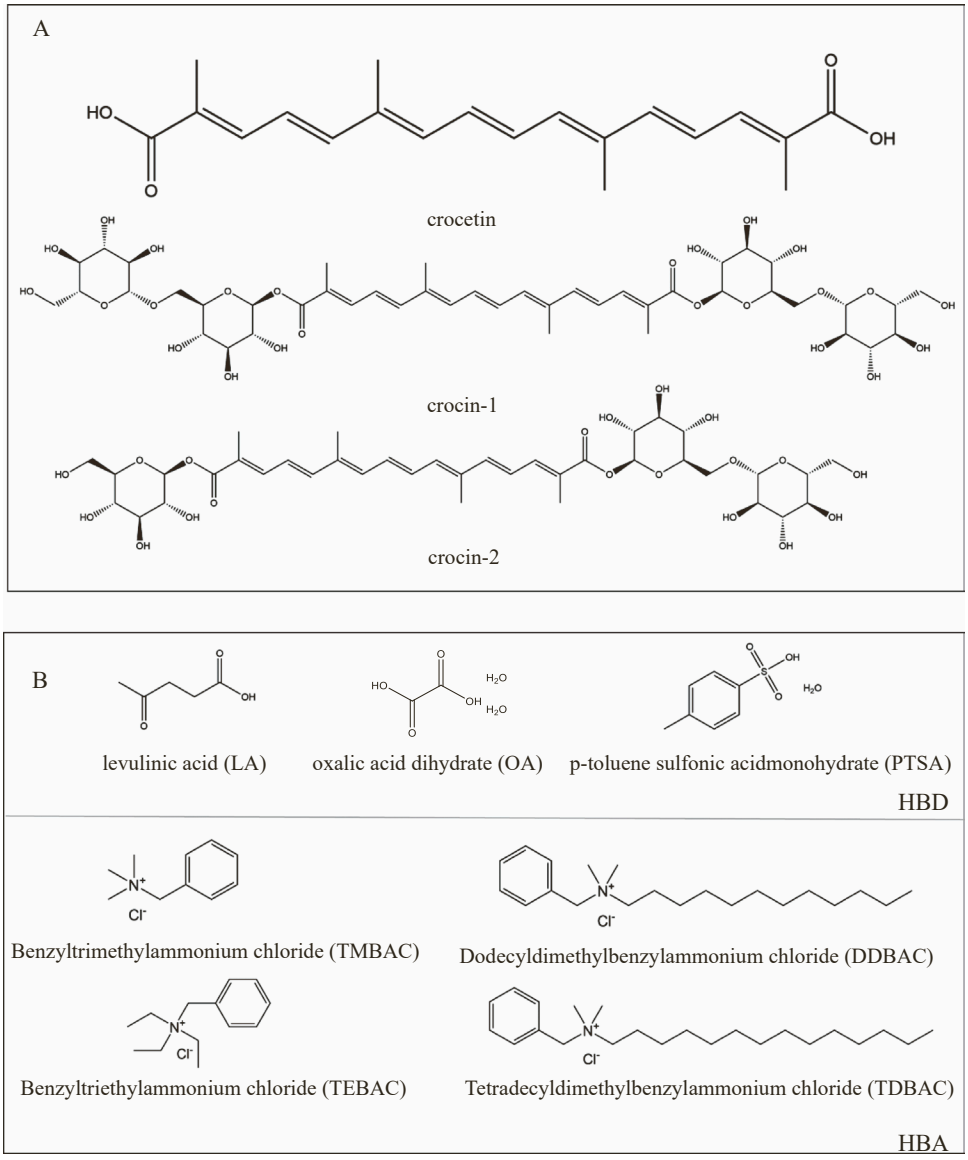


Fig. 1. (A) The structures of crocetin and crocins; (B) The structures of seven compounds used for the preparation of deep eutectic solvents.

bond acceptor (HBA) and hydrogen bond donor (HBD) molecules in specific molar ratios, can achieve viable structural configurations through the manipulation of HBD and HBA combinations (Liu et al., 2022). In addition, the remarkable characteristic of DES lies in their compatibility with biomolecules, rendering them suitable for bio-organic catalysis, bio-transformation, and molecular extraction within the biopharmaceutical industry (Satlewal et al., 2018). By designing a task-specific system, the target product can be dissolved in it, while the impurities are insoluble. Apart from its role as a solvent, DES could also participate in synthesizing materials by decomposition (Yu et al., 2021). In general, DES have exhibited considerable potential as environmentally friendly solvents for extracting bioactive constituents from herbaceous plants (Huang et al., 2022; Mansur et al., 2019).

Building upon the utilization of ionic liquids as dual solvent-catalysts, it is hypothesized that the deglycosylation process of crocin to obtain crocetin could be accomplished in DES without the need of enzymes (Kalyniukova et al., 2021). A series of hydrophilic eutectic solvent systems were designed and thoroughly characterized to produce crocetin from *Gardenia jasminoides* Ellis. And the Box-Behnken Design (BBD)-response surface method (RSM) was used to optimize processing conditions (Zhang et al., 2022). Moreover, the mechanism of simultaneous extraction and conversion was elucidated. In this study, a straightforward and convenient method was employed to recover crocetin after extraction, and the potential of the recovered DES for recycling was assessed. Consequently, a feasible and environmentally sustainable technique for the one-step production of crocetin from gardenia fruit was developed.

2. Materials and methods

2.1. Chemicals and reagents

Crocetin (98%), levulinic acid (LA, 98%), p-toluene sulfonic acid monohydrate (PTSA, 99%), oxalic acid dihydrate (OA, 99%), benzyltrimethylammonium chloride (TMBAC, 98%), benzyltriethylammonium chloride (TEBAC, 99%), dodecyldimethylbenzylammonium chloride (DDBAC, 99%), tetradecyldimethylbenzylammonium chloride (TDBAC, 98%), α-crocin (reference substance), sodium hydroxide (NaOH, 95%), were obtained from Shanghai Macklin Biochemical Technology Co., Ltd. 2,2-Diphenyl-1-picrylhydrazyl (DPPH) was purchased from Sigma-Aldrich. The HPLC-grade acetonitrile and acetic acid were bought from Meridian Medical Technologies Ltd. The analytical grade methanol, ethyl acetate, n-butanol, hexane and dimethyl sulfoxide were bought from Beijing Chemical Works. Pure water was manufactured by a Milli-Q water purification system. *Gardenia jasminoid*e Ellis was supplied from Beijing Tong Ren Tang Technology Development Co., Ltd. Human immortal skin keratinocytes (HaCaT), bought from Cell Resource Center, IBMS, CAMS/PUMC. The Dulbecco's modified Eagle's medium (DMEM), foetal bovine serum (FBS), trypsin/ethylenediaminetetraacetic acid (EDTA) solutions (0.25%) and antibiotics (penicillin and streptomycin sulfates) were purchased from Gibco. All the cell culture ingredients and phosphate buffered saline (PBS) were purchased from Invitrogen. Antibiotic solutions and (3-(4,5-dimethylthiazol-2-yl)− 2,5-diphenyltetrazolium bromide (MTT) were purchased from Biotopped Science & Technology CO., Ltd. Sodium chloride, isobutanol and hydrochloric acid were purchased from Sinopharm Chemical Reagent Co., Ltd.

2.2. Preparation and characterization of deep eutectic solvents

Levulinic acid, p-toluene sulfonic acid monohydrate, and oxalic acid dihydrate were selected as HBDs, and various benzylamine chloride species with different alkyl substituents were chose as HBAs. Their structures are shown in Fig. 1(B). All eutectic mixtures were heated at 70 °C and stirred for one hour until a stable and transparent liquids were formed. The melting points of different DES systems were investigated by a differential scanning calorimeter (DSC, TA Instruments Q2000, USA, accuracy ± 0.1 °C, precision ± 0.02 °C, ranging from −100 to 200 °C). To ensure a continuous flow of nitrogen gas as the purifying gas, a heating/cooling rate of 10 °C/min was used. All eutectic systems and their pure components were measured using attenuated total reflection Fourier-transform infrared spectroscopy (ATR-FTIR, IRTracer-100, SHIMADAU, Japan, absorption mode, 50 scans, ranging from 4000 to 400 cm^{-1}). The DES and its recycled were characterized by NMR. An Agilent DD2 operating at 600 MHz was used at 25 °C with deuterated water as the solvent. Thermogravimetric analysis (TGA) was performed on a NETZSCH STA 449F5 synchronous thermal analyzer manufactured by Netzsch (Germany), in the temperature range 40 °C–600 °C, heating rate 10 °C/min under nitrogen atmosphere. The pH values of DES were estimated at 80 °C using a pH meter (PHSJ-5, INESA, China). The acidity of DES was determined as described in previous literature (Jablonský and Jančíková, 2023). Firstly, DES (1 g) was mixed with 20 mL of deionized water. Subsequently, this prepared mixture was titrated with a NaOH solution with a concentration of 0.5 N until pH = 7 (equivalence point) was reached. The Kamlet-Taft solvatochromic parameters for DES were carried out according to the published protocols (Zhang et al., 2023; Dwamena and Raynie, 2020). The structures of these mixtures and target were further confirmed by ^{1}H nuclear magnetic resonance (^{1}H NMR) and ^{13}C nuclear magnetic resonance (^{13}C NMR) spectroscopy at 600 MHz (Agilent DD2 spectrometer, Germany). Thermogravimetric analysis (TGA) of system was performed with a SDT Q600 analyzer from 25 to 600 °C with a step rate of 10 °C/min.

2.3. Production of crocetin from gardenia fruit

DES (900 μL) and gardenia fruit powder (0.15 g) were placed in a polytetrafluoroethylene (PTFE) centrifuge tube and then shaken in a constant-temperature metal bath (JXH-100, China) at 800 rpm and 70 °C for 30 min throughout the screening procedure of different solvent systems. Following the screening of DES, the selected DES was used to optimize simultaneous extraction and conversion process. Single-factor experiments were conducted to study the effect of five factors, including mole ratio of HBA to HBD (2.5:1, 2:1, 1.5:1, 1:1, 1:1.5), the ratio of powder weight to solvent volume (1:6, 1:10, 1:20, 1:30, 1:40), processing temperature (40–90 °C) and time (10–100 min). All tests were performed in three parallel samples and were analyzed by HPLC. The method of targets analysis and determination of antioxidant capacity can be found in the supplementary material. A model was used to reveal the conversion of crocin to crocetin in DES following the reference (Yu et al., 2023), which was expressed by the following Eq. (1). Langmuir-Hinshelwood model (Khan et al., 2021) was used to investigate the relationship between sample concentration and crocetin yield.

$$C = C_e - C_e e^{-kt} \quad (1)$$

where C is crocetin yield (mg/mL), Ce is crocetin yield in the equilibrium state (mg/mL), k is reaction rate constant (min^{-1}), and t is reaction time (min).

2.4. Response surface optimization

All experiments were performed in triplicate, and results were presented as means ± standard deviations (SD). One-way analysis of variance (ANOVA) was applied and factors with significant differences were chosen to be optimum by BBD in response surface methodology (RSM) using Design Expert 10.0. Based on the findings of the one-single factor experimental results, three variables were chosen for the production of crocetin: ratio of solid to liquid (mL/g), processing temperature (°C), and processing time (min). The ideal findings of one-single factor experiment were used to determine the ranges of solid-liquid ratio (A), processing temperature (B), and processing time (C). The experimental data were fitted to the following second-order polynomial regression

model (2):

$$Y = \beta 0 + \sum_{i=1}^{k} \beta iXi + \sum_{i=1}^{k} \beta iiXi^2 + \sum_{i=1}^{k} \beta ijXiXj \quad (2)$$

where Y is the response variable, X_i and X_j are the independent variables, and k denotes the number of tested variables (k = 3). β_0 signifies that the intercept is a constant term, and β_i, β_{ii} and β_{ij} are the linear, quadratic, and cross-product terms, respectively.

2.5. Recovery of DES and crocetin

The processing solution was diluted with a substantial volume of water before being passed through a 0.2 μm membrane. At this moment, DES was dissolved in water and flowed through the membrane, while crocetin precipitated and failed to pass through the membrane, yielding crude product of crocetin. To recover the DES, an aqueous solution of DES was subjected to extraction using a modest quantity of n-butanol to eliminate the remaining impurities. The DES was recovered by removing the water through rotary evaporation. Following this, the crude product of crocetin underwent further refinement. Most impurities were initially eliminated by washing with a small amount of methanol, followed by a 50% (v/v) aqueous solution of dimethyl sulfoxide and methanol. Subsequently, the crocetin crystals were washed with a solution of ethyl acetate-hexane (2:8, v/v) and a 50% (v/v) aqueous methanol solution to obtain crocetin crystals of high purity after a substantial amount of water was added to the crocetin-dimethyl sulfoxide solution. The flowsheet depicting the concurrent extraction and conversion process utilizing the proposed DES, and the recovery process of crocetin and DES is presented in Fig. 2.

2.6. Cytotoxicity on human keratinocytes

The human keratinocytes (HaCaT) cells were cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and 1% antibiotics (100 U/mL penicillin and 100 mg/mL streptomycin sulfates) at a temperature of 37 ℃ in a humidified incubator with 5% (v/v) CO_2. The medium was changed every two days, and the cells were treated with a 0.25% trypsin/ethylenediaminetetraacetic acid (EDTA) solution. Cell viability was assessed using the 3-[4,5]dimethylthiazol-2,5dephenyltetrazolium bromide (MTT) assay, which measured the reduction of the soluble yellow MTT tetrazolium salt to its blue insoluble MTT formazan product. HaCaT cells were seeded at a density of 1×10^4 cells per well in 96-well plates and cultured until reaching 80% confluence for the purpose of conducting a cytotoxicity assay. Subsequently, the cells were exposed to varying concentrations of DES for a duration of 24 h. Following this incubation period, each well was treated with 10 μL of MTT (5 mg/mL) and incubated at a temperature of 37 °C for a duration of 4 h. The resulting formazan product was then dissolved in a lysis solution consisting of 10% sodium dodecyl sulfate, 5% isobutanol, and 12 mM HCl in H_2O. Absorbance at a wavelength of 570 nm was measured using a Tecan Infinite M200 Pro multimode microplate reader (Tecan Trading AG, Switzerland). The cell viability was calculated as a percentage of the vehicle control. Likewise, IC_{50} (half maximal inhibitory concentration) was defined as the chemical concentration that causes a 50% drop in cell viability, as estimated using a logistic function (non-linear regression).

3. Results and discussion

3.1. Characterization of deep eutectic solvents

The prepared eutectic mixtures underwent characterization through DSC and NMR. Fig. 3(A) reveals the identification of a novel DES, which exhibits a significantly reduced melting point compared to the individual components due to their strong and specific interaction. Furthermore, the chemical signals of the H atoms in TEBAC shift to a higher field, as seen in Fig. 3(B) (^{1}H NMR), indicating the presence of cluster aggregates created by OA surrounding TEBAC. Additionally, FTIR spectroscopy was employed to ascertain the chemical structures of p-toluene sulfonic acid monohydrate, levulinic acid, oxalic acid dihydrate, TEBAC, and the proposed DES to confirm the presence of bonds in individual components as well as noncovalent interaction between HBD and HBA in DES (Fig. S1). These characterizations illustrated the successful formation of eutectic solvent systems (Mat Hussin et al., 2020). As k and E_a kinetic parameters of DES cannot be obtained by isothermal methods like ionic liquids, the onset temperature of degradation/mass loss (T_{onset}) and the temperature of maximum weight loss rate (T_{peak}) were used to evaluate the short-term stability of DES, which could be investigated by fast scan TGA (Chen et al., 2018). The T_{onset} and T_{peak} values of TEBAC-OA (1:1.5) were found to be 120.1 °C and 158.4 °C (Fig. S2), indicating that this system has moderate thermal stability and may find application in processes requiring somewhat higher heating temperatures (Chen et al., 2023a, 2023b). Therefore, it would remain stable during the proposed operation at a relatively high temperature.

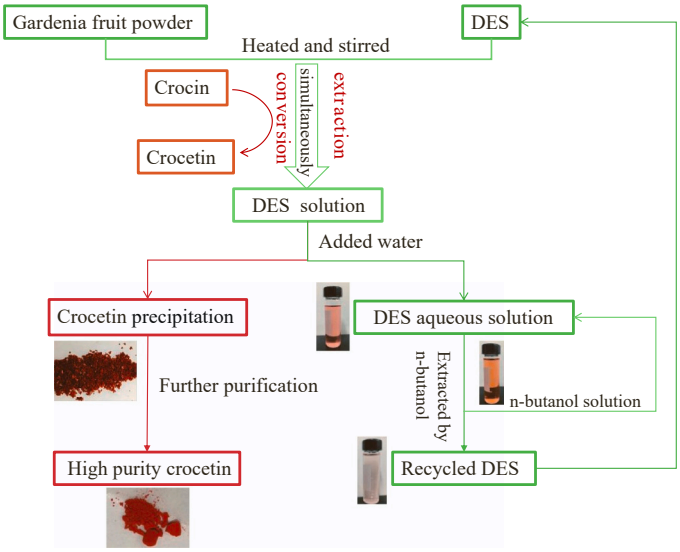


Fig. 2. Flowsheet of the simultaneous extraction and conversion process using the proposed deep eutectic solvent, and recovery process of crocetin and DES.

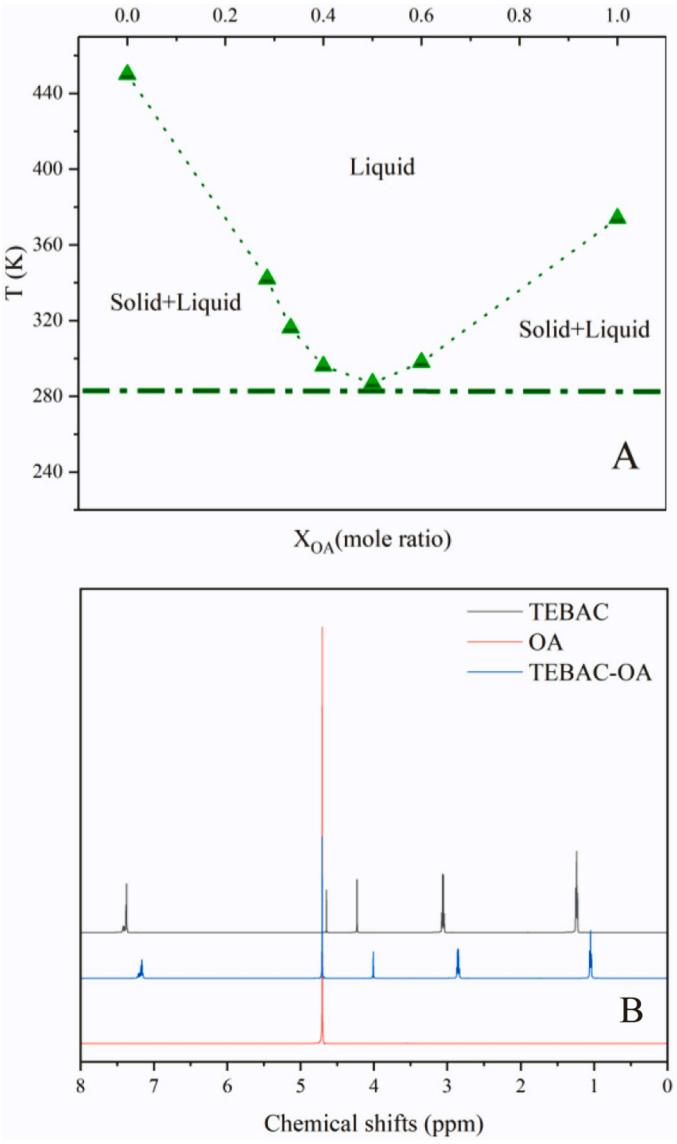


Fig. 3. Characterization of deep eutectic solvent. (A) Experimental solid-liquid equilibrium phase diagram of TEBAC-OA solvent system; (B) ^{1}H NMR spectra of triethylbenzylammonium chloride-oxalic acid dihydrate (TEBAC-OA: 1:1.5) and its components.

3.2. Screening of deep eutectic solvent

As previously reported, the concentration of crocetin in collected gardenia fruit ranges from 0.004% to 0.025% (Chen et al., 2011). The initial crocetin content in the plant was determined based on the optimal conditions established in previous studies for extracting bioactive compounds. These conditions included an ethanol concentration of 51.3%, a solid to liquid ratio of 1:6 (g/mL), a processing temperature of 70.4 °C, and a duration of 28.6 min in a water bath (Yang et al., 2009). In this study, TEBAC was initially selected as the HBA to investigate the impact of various HBDs. Eutectic systems with an HBA:HBD ratio of 1:1.5 were prepared using oxalic acid, p-toluenesulfonic acid, and levulinic acid as HBDs. The yield of crocetin utilizing these DES is shown in Fig. 4(A). The superior performance of the oxalic acid-based system is evident. We have determined the acidity of these solvent systems, as illustrated in the supplementary material. Although the disparity was not substantial, it was worth noting that the TEBAC-OA solvent system, which exhibited the highest target yield, also displayed the highest acidity among the three proposed solvent systems. This observation suggests that the HBD component exerts a significant influence on the acidity of DES and its overall properties. The Kamlet-Taft solvatochromic parameters provided additional confirmation of the aforementioned findings. Specifically, the α value, which indicated the hydrogen-bond-donating ability (i.e., Lewis acidity), of TEBAC-OA was significantly higher compared to the other solvents. Addtionally, the yield of crocetin was directly proportional to the α value. Furthermore, the optimization of the chain length of the HBA in the oxalic acid-based system was conducted. As illustrated in Fig. 4(B), the highest yield of crocetin was achieved by the DES composed with TEBAC. Subsequently, the molar ratios of HBD and HBA were optimized. As shown in Fig. 4(C), the yield exhibited a decreasing trend with an increase in the molar ratio of HBD and HBA. This phenomenon can be attributed to the hydrogen-bond interactions between crocetin and oxalic acid. An increase in the oxalic acid ratio would result in a greater number of molecules capable of interacting with both the DES and crocetin, thereby improving the efficiency of production. However, it should be noted that the formation of the DES was no longer possible once the HBA:HBD ratio exceeded 1:1.5. Consequently, the DES known as TEBAC-OA (1:1.5) was chosen for further utilization in subsequent optimization processes.

3.3. Optimization of simultaneous extraction and conversion conditions

Additionally, selecting an appropriate solid-liquid ratio can not only enhance the production of the desired chemical but also minimize solvent consumption, leading to cost savings (Su et al., 2017). To enhance the production of crocetin, the simultaneous extraction and conversion process was carried out using TEBAC-OA (1:1.5) as the extracting agent at various solid-liquid ratios (1:6, 1:10, 1:20, 1:30, and 1:40 g/mL). As shown in Fig. 4(D), The processing efficiency is enhanced by augmenting the solid-liquid ratio from 1:6 to 1:20. However, as the solid-liquid ratio continues to rise, production efficiency exhibits a declining trend. This can be attributed to the potential solubilization and disintegration of cell walls, which may lead to an increased overflow ratio of the desired content. Nevertheless, excessive solubility can result in a saturation phenomenon, causing certain components to remain undissolved. Consequently, this leads to wastage of the solvent and complicates the process, ultimately impacting the overall effectiveness (Wang and Li, 2022). Therefore, the solid-liquid ratio of 1:20 was selected.

Subsequently, the processing temperature was varied within the range of 40 to 90 °C, with a 10 °C increment (Fig. 4(E)). The yield of crocetin exhibited a significant increase from 40 to 80 °C, followed by a decline from 80 to 100 °C. This phenomenon can be attributed to the fact that raising the temperature of the DES generally leads to a reduction in viscosity and surface tension, as well as an enhancement in the diffusion of targets. These changes facilitate the permeation of DES into the plant material matrices, enabling the desorption and dissolution of the desired compounds into the solvent, thereby augmenting the conversion yield. However, while an elevated processing temperature may enhance the yield, it may not effectively mitigate the thermal degradation of active compounds at higher temperatures (Xu et al., 2017). When the temperature of the solvent system surpassed 80 °C, the structural stability of crocetin decreased, leading to a subsequent decline in production rate. Previous research has taken advantage of crocin's substantial solubility in DES to achieve high extraction efficiencies (Wen et al., 2023). Considering the aforementioned characteristics, particular emphasis should be placed on the deglycosylation effects when employing the suggested bifunctional DES (Chen et al., 2023a, 2023b).

The effect of processing time in the range of 10 to 100 min was investigated using TEBAC-OA (1:1.5) as the solvent at 80 °C. As shown in Fig. 4(F), the yield of crocetin maintains growth until 30 min, after which it begins to decline, likely due to the degradation caused by the heat instability of crocetin. Typically, the active components cannot be removed entirely in a short period of time. Prolonged processing can affect the stability of the material, particularly the activity of DES,

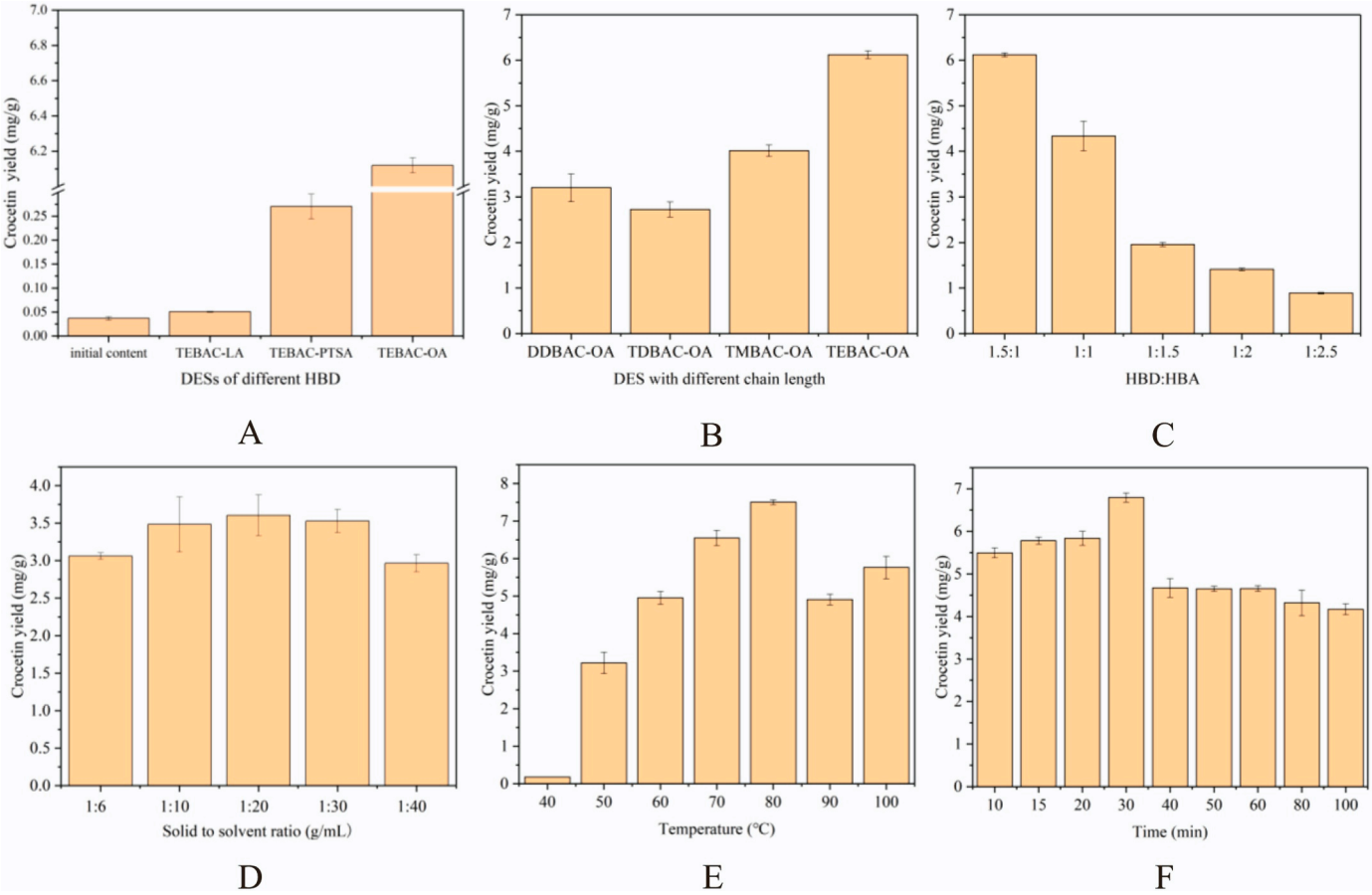


Fig. 4. The effect of different hydrogen bond donor (A); different hydrogen bond acceptor (B); different molar ratio of hydrogen bond donor to hydrogen bond acceptor (C); different solid-liquid ratio (D); temperature (E) and time (F) on the yield of crocetin. TEBAC-PTSA (triethylbenzylammonium chloride-p-toluene sulfonic acidmonohydrate); TEBAC-LA (triethylbenzylammonium chloride-levulinic acid); TEBAC-OA (triethylbenzylammonium chloride-oxalic acid dihydrate); DDBAC-OA (dodecyldimethylbenzylammonium chloride-oxalic acid dihydrate); TMBAC-OA (benzyltrimethylammonium chloride-oxalic acid dihydrate); TDBAC-OA (tetradecyldimethylbenzylammonium chloride-oxalic acid dihydrate);.

consequently impacting production efficiency (Cao et al., 2018). Consequently, 30 min was determined to be the ideal processing time. The role of water in DES can vary depending on its concentration. As anticipated, the reaction rate gradually diminishes with increasing water content (Fig. 5(A)). It has been estimated that the presence of water leads to a decrease in the robust hydrogen bond networks, consequently diminishing the solvent's ability to catalyze crocins (Chen et al., 2023a, 2023b). The acidity of DES with varying water contents was also assessed (supplementary material). The acidity of the solvent decreased with the increase of water content in the system, and the yield of crocetin exhibited a positive correlation with the acidity. This serves as further evidence supporting the role of acidity in the reaction. There existed a balance between water content and solvent acidity. A portion of crocins was extracted into the eutectic system and instantly catalyzed

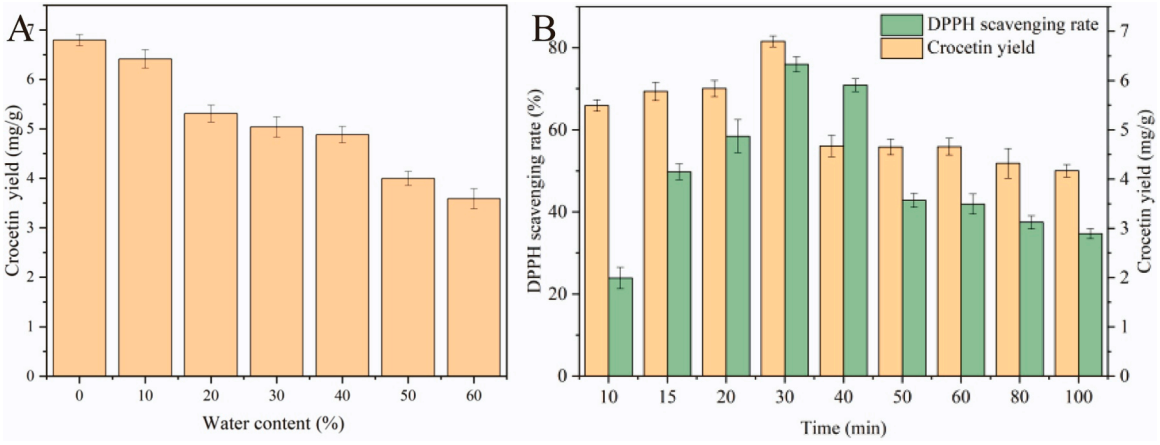


Fig. 5. (A) The effect of different water content on the yield of crocetin; (B) The effect of different processing time on the antioxidant activity of DPPH (left ordinate) and the yield of crocetin (right ordinate).

to crocetin. The crocins in DES remained in an unsaturated state, thus the catalytic conversion ability of the DES was considered as the main factor that affects the yield of crocetin. The internal bound waters in the proposed solvent system or the plant raw materials were sufficient to promote the conversion reaction. Consequently, the eutectic solvent system devoid of water was suitable for the concurrent extraction and conversion procedure. Meanwhile, the DPPH antioxidant activity exhibits a positive correlation with the crocetin production rate, as depicted in Fig. 5(B). Notably, the highest antioxidant activity occurs after 30 min of processing time, which aligns with the duration yielding the maximum output, thereby providing the evidence of the antioxidant activity of crocetin.

3.4. Using Box-Behnken design to optimize parameters

The independent variables, along with their respective codes and levels, are presented in Table S1. The experimental design consisted of 17 randomized runs, as outlined in Table S2. By analyzing the experimental data, the regression Eq. (3) was obtained as follows:

$$Y = -44.47641 + 0.26572A + 0.93041B + 0.75611C + 0.0000722305A * B - 0.000920438A * C - 0.00431955B * C - 0.00412416A^2 - 0.00535856B^2 - 0.00651521C^2 \quad (3)$$

The outcomes of the fitted response surface model, following a significance test, are displayed in Table S3. The analysis demonstrated that the model was statistically significant, with an F-value of 56.65 ($P < 0.0001$). The "Lack of Fit F-value" of 3.61 indicated that the discrepancy between the fitting term and pure error was not significant ($p = 0.123$), thus confirming the validity of the model. The determination coefficient (R^2) yielded a value of 0.9865, signifying that the fitted model had the capacity to account for 98.6% of the observed variations. Similarly, the adjusted determination coefficient (R^2_{Adj}) attained a value of 0.969, which closely resembled R^2, thereby indicating a strong correlation between the observed and predicted values. Moreover, the low coefficient of variation (CV = 7.88%) indicated minimal fluctuations in the mean value, suggesting the reliability of experimental results with a high degree of precision. Collectively, these findings substantiate the accurate depiction of the true relationship between the response and the independent variables by the model. The degree of interaction between the two parameters positively correlates with the magnitude of slope change observed on the 3D surface, the density of contour lines, and the resemblance to an oval or saddle shape. Fig. 6 illustrates that the contour lines representing the BC, AC, and AB interaction terms predominantly exhibit an elliptic shape. The inclination angle of the surface is found to be favorably influenced by the interplay between the two factors. A comparison suggests that the impact of the interaction between the two factors follows the pattern: processing temperature and processing time (BC) >solid-liquid ratios and processing time (AC) > solid-liquid ratios and processing temperature (AB). This finding aligns with the analysis of variance results presented in Table S3. Based on the regression equation and the results of variance analysis, the optimal processing conditions were obtained. The model optimized the solid-liquid ratio (A) at 1:30.053 (g/mL), the processing temperature (B) at 79.374 °C, and the processing time (C) at 30.499 min. The maximum yield was estimated to be 8.541 mg/g under these optimal circumstances. Validation experiments were conducted under ideal conditions to verify the accuracy of the predicted values. These experiments resulted in a yield of 8.485 mg/g, which closely aligned with the predicted value from the model.

3.5. Process scale-up and recovery of crocetin and DES

The processing solvent used in this study was TEBAC-OA (1:1.5), and the volume of the processing solvent was increased by a factor of 3–36 under the optimal conditions, which included a solid-liquid ratio of 1:30 (g/mL), a processing temperature of 80 °C, and a processing time of 30 min. The results presented in Fig. S3 demonstrate that the production efficiency remains nearly consistent with small-scale quantities when the processing scale is increased, indicating the feasibility of mass manufacturing. However, the non-volatility of DES poses significant challenges to the efficient and rapid recovery of target compounds from DES extracts (Cai et al., 2021). Nevertheless, the recycling and reuse of DES are essential and crucial for achieving environmentally friendly and cost-effective production of valuable compounds. In this study, crocetin was successfully recovered by diluting it with water, resulting in a remarkable recovery rate of 94%. The recovered DES was then utilized in the subsequent processing cycle, yielding 8.15 mg/g, which accounted for 95.8% of the yield obtained with fresh DES. Furthermore, the reusable performance of the used DES was further assessed. During the fourth cycle, the efficiency of DES was 70.8% of the fresh DES, indicating a limit on the times of reusable cycles (Fig. S4). Moreover, ^{1}H and ^{13}C NMR were used to investigate the structure of the used DES and purified DES (Fig. S5). There were trace impurities, whose chemical shifts ranged from 7 to 8 ppm. They possibly originated from sugar or other water-soluble substances derived from the plant material. After purifying the crude crocetin product, highly pure trans-crocetin crystals were obtained. The identification of the purified crocetin was confirmed through ^{1}H NMR and ^{13}C NMR spectral data, which were compared with the reference (Table S4), and the spectra were depicted in Fig. S6, demonstrating consistency with the reported literature (Wang et al., 2021). All in all, the utilization of the proposed DES enabled the attainment of crocetin within a mere 30 min via a single step, and the recovery of crocetin necessitated solely the addition of water. Although n-butanol was required to purify the DES, both n-butanol and DES could be reused. The corresponding investigation of mechanism can be found in the supplementary material (Table S5-Table S7, Fig. S7- Fig. S10), and the acidity of eutectic systems greatly affected the yield of the product.

3.6. Cytotoxicity of DES

The HaCat cell line was employed to assess the biological toxicity of the proposed solvent systems for simultaneous extraction and conversion. Specifically, the cytotoxicity of four levulinic acid-based DES could be ranked as follows: TMBAC < TEBAC < DDBAC < TDBAC, indicating that the toxicity of levulinic acid-based systems was significantly influenced by the alkyl chain length of HBA, which is fundamentally positively associated, corresponding with the assumptions made in the existing literature (Stolte et al., 2007). The presence of a longer alkyl chain disrupts the integrity of the cellular membrane, leading to the development of cytotoxic effects. Additionally, the acid HBDs exhibited relatively minor impacts on cytotoxicity, with fewer contributions compared to HBAs of varying carbon lengths: oxalic acid < p-toluene sulfonic acid < levulinic acid. The cytotoxicity of oxalic acid-based eutectic systems showed the same tendency. Furthermore, the toxicity of these DES was compared to other eutectic systems (Macário et al., 2019), ionic liquids (Zanoni et al., 2019), and conventional solvents (Ilieva et al., 2021) reported in the literatures. More importantly, as indicated in Table 1, DES containing TEBAC and TMBAC exhibited significantly lower cytotoxicity compared to other ammonium salt-based ionic liquids, ammonium chloride-based DES, and traditional organic solvents. The cytotoxicity tests further confirmed the safety and environmentally friendly nature of the proposed DES, which was utilized as the processing solvent in this study.

4. Conclusion

In this study, a series of benzylammonium chloride-based DES were prepared and characterized using FTIR, DSC, and cytotoxicity analysis. The eutectic mixtures, which consist of organic acids and amine salts, can hydrolyze the glycosidic bond of crocins, thereby facilitating solubilization and deglycosylation. This approach overcomes the limitations associated with other methods, such as prolonged reaction times,

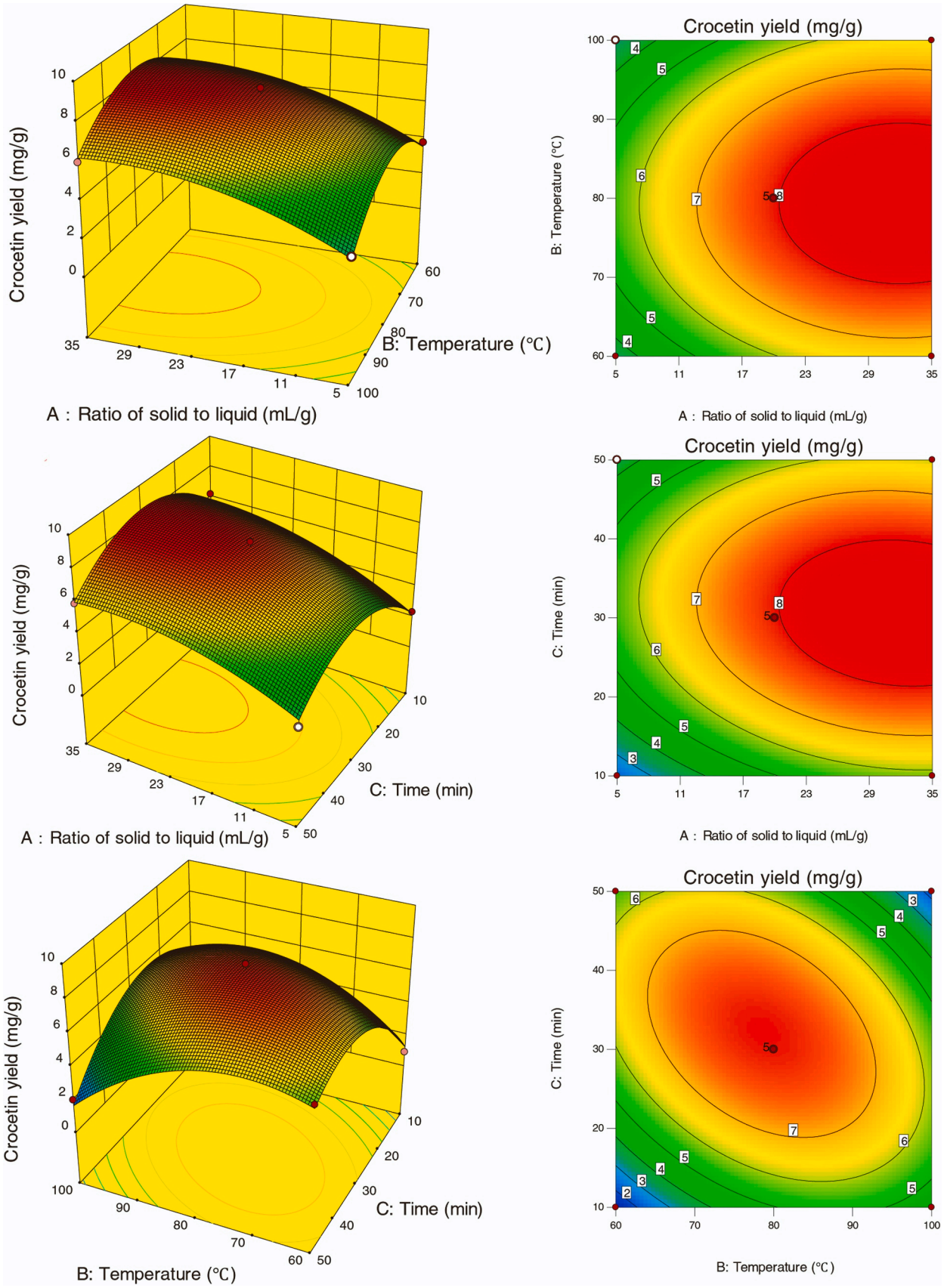


Fig. 6. Optimization of the yield of crocetin using Box–Behnken Design (BBD).

Table 1
Comparison of the cytotoxic effects of the proposed DES with other solvents.

	Solvent	Cell line	IC_{50} (μg/mL)
DES (1:2)	levulinic acid-DDBAC	HaCat	5.9
	levulinic acid-TDBAC	HaCat	5.1
	levulinic acid-TMBAC	HaCat	3418.1
	levulinic acid-TEBAC	HaCat	2546.7
	oxalic acid dihydrate-TEBAC	HaCat	3336.5
	p-toluene sulfonic acid monohydrate-TEBAC	HaCat	33132
	oxalic acid dihydrate-TMBAC	HaCat	8638.2
	oxalic acid dihydrate-DDBAC	HaCat	1.7
	oxalic acid dihydrate-TDBAC	HaCat	4.7
DES (Macário et al., 2019)	tetrabutylammonium chloride-Butanoic acid	HaCat	108.7
	tetrabutylammonium chloride-Hexanoic acid	HaCat	112.0
	tetrabutylammonium chloride-Ethylene glycol	HaCat	34.1
	tetrabutylammonium chloride-1-Propanol	HaCat	77.9
	tetrabutylammonium chloride-Urea	HaCat	81.8
Ionic liquids (Zanoni et al., 2019)	2-Hydroxy ethylammonium propionate	HaCat	9.0
	2-Hydroxy diethylammonium lactate	HaCat	17.3
	2-Hydroxy diethylammonium benzoate	HaCat	12.4
	2-Hydroxy diethylammonium salicylate	HaCat	11.7
	2-Hydroxy diethylammonium adipate	HaCat	26.9
Traditional solvent (Ilieva et al., 2021)	DMSO	HaCat	170.0
	Ethanol	HaCat	190.0

excessive reagent consumption, and residual solvents. Through the implementation of a one-single factor experiment and response surface methodology, TEBAC-OA (1:1.5) was specifically tailored for the desired processing conditions, with a solid-to-liquid ratio of 1:20 and heating at 80 °C for 30 min. The yield of crocetin was found to be 8.485 mg/g. Following the processing, DES and crocetin were successfully recovered separately, resulting in a 94% recovery rate for crocetin. The used DES was then utilized in the subsequent processing step, maintaining a production efficiency of 95.8%. The effective recovery of DES demonstrated the repeatability, environmental friendliness, and cost-effectiveness of this processing method. Additionally, the cytotoxicity assessment revealed that the final used eutectic systems exhibited lower cytotoxicity compared to other commonly used solvents. The hydrogen bonding energy between DES and crocin, as well as the acidity of DES, were found to have a strong correlation with the yield of crocetin. This paper not only provides a sustainable, efficient, and simple method for the one-step production of crocetin, but also broadens the application of DES in processing, which provides new strategies and ideas for the application of DES in plant-based raw materials.

CRediT authorship contribution statement

Wen Lijiao: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Fan Chen:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization, Project administration, Resources, Visualization. **Cao Xueli:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data that has been used is confidential.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2023.117969.

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